

Bandwidth-Extended High-Efficiency Rectifier Using Shunt and Serial Impedance Compensation

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Abstract—This letter presents a compact microwave rectifier using a novel impedance matching technique to extend the operating bandwidth. Originating from conducting impedance controls at the edge and central part of the targeted frequency band, the matching network can manipulate the impedance over a wide frequency band. By paralleling an open-circuit stub and then connecting a capacitance in series, the impedance variation in the rectifying diode can be significantly reduced. Theoretical analysis and simulations of the proposed structure were carried out. A prototype was fabricated, which demonstrates a broadband bandwidth of 51.9% (2.5–4.25 GHz) where the RF–dc conversion efficiency is higher than 70% at 11 dBm, and $|S_{11}|$ is also less than -10 dB within this frequency range.

Index Terms—Broadband impedance matching, energy harvesting (EH), microwave rectifier, wireless power transfer (WPT).

I. INTRODUCTION

WIRELESS power transfer (WPT) and energy harvesting (EH) are promising microwave/RF technologies that offer to extend the battery life of a device and allow applications where wired power is hard to access or unavailable [1], [2]. Recently, broadband EH is becoming popular where a broadband rectifier is needed [3]. Besides, a WPT system is typically integrated with other high-speed communication systems, such as simultaneous wireless information and power transfer (SWIPT) [4] and wireless sensor networks [5], [6]. Hence, it is expected to have a rectifier with both a high conversion efficiency and a wide bandwidth [7], [8].

Several research groups offered different design architectures to improve the efficiency of a rectifier or rectenna over a broad bandwidth [9], [10]. Bolos et al. [11] introduced a one-octave bandwidth rectifier that is based on a nonuniform transmission line. In this design, although the efficiency (η) remains above 60% from 470 to 860 MHz at 10 dBm, the nonuniform transmission line introduces additional insertion

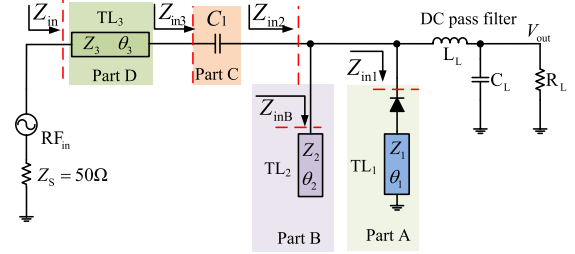


Fig. 1. Schematic of the proposed broadband rectifier. Parts B and C are used for shunt and serial impedance compensation, respectively.

loss, and meanwhile considerably increases the physical size of the rectifier. In [12], a fractional bandwidth of 21.5% ($\eta > 70\%$) was realized by a second-order branch-line coupler at the input. It is noted that the coupler also introduces an additional insertion loss, which compromises the efficiency.

Designing a high-efficiency broadband rectifier using a single diode presents significant challenges due to the diode's impedance is more sensitive to frequency when compared with a voltage doubler [13]. Moreover, a matching network with low insertion loss and compact size is essential. In [14] and [15], a shunt stub with odd-symmetric admittance characteristics at the center frequency can effectively neutralize the impedance variations of the circuit to be matched. Taking inspiration from this topology, this letter demonstrates that after compensation with a shunt resonator, the bandwidth can be further broadened by incorporating a series capacitor, which also serves as a dc-blocking capacitor.

II. DESIGN OF THE MICROWAVE RECTIFIER

Fig. 1 presents the schematic of the proposed single-shunt rectifier with a low-loss matching network consisting of Parts A–D, and the input impedances associated with them are denoted as Z_{in1} , Z_{in2} , Z_{in3} , and Z_{in} respectively. In the following analysis, we define the lower and upper frequencies of the operating band as f_L and f_H , respectively, and the frequency ratio k is defined as f_H/f_L where $f_L = 2.5$ GHz.

A. Design of Part A

As indicated in Fig. 2(a), Part A is a short-ended stub which is in serial with the diode, and it has a characteristic impedance Z_1 and an electrical length θ_1 at f_L . It is designed to compensate the capacitive impedance of the diode, and after that, the imaginary part of Z_{in1} at f_L and f_H has equal magnitude but opposite signs.

Closed-form equations for the diode's efficiency and input impedance have been derived in [16]. Using those equations, the diode impedance can be modeled as a parallel connection of a nonlinear junction capacitance C_e and a junction resistance R_e [see Fig. 2(a)]. Assuming the rectifier maintains a stable efficiency for a given input power P_{in} and dc load, C_e

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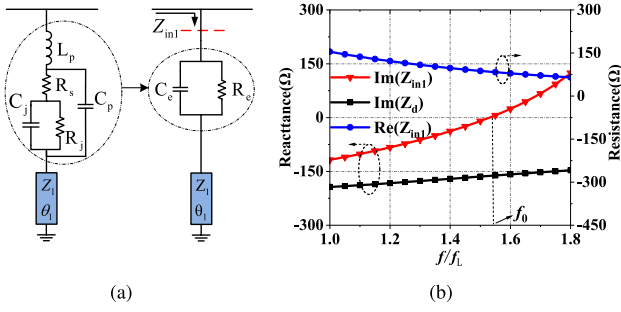


Fig. 2. (a) Schematic of Part A and the equivalent circuit model of the diode. (b) Calculated diode impedance and Z_{in1} after being offset by Part A.

and R_e can be regarded as constant. So, as shown in the right part of Fig. 2(a), Z_{in1} with frequency can be expressed as

$$Z_{in1}(f) = \frac{R_e}{1 + (R_e \omega C_e)^2} - j \frac{R_e C_e \omega}{1 + (R_e \omega C_e)^2} + j Z_1 \tan \theta_1(f). \quad (1)$$

The $\theta_1(f_L)$ and $\theta_1(f_H)$ are the electric lengths at f_L and f_H , respectively, so we have

$$\theta_1(f_H) = k \theta_1(f_L) = k \theta_1. \quad (2)$$

To maximize the operational bandwidth, a target optimization method for frequency ratio k is adopted. k is optimized by tuning the characteristic impedance Z_1 (20–120 Ω) and electrical length θ_1 (0°–180°) of transmission line TL_1 . The solution space is constrained by imposing

$$\text{Im}[Z_{in1}(f_L)] + \text{Im}[Z_{in1}(f_H)] = 0 \quad (3)$$

at band-edge frequencies $\{f_L, f_H\}$.

For an HSMS-286 diode, when the RF–dc efficiency is 0.6 and the input power is 0 dBm, C_e and R_e were extracted as 0.23 pF and 400 Ω, respectively. Using (3), the optimal parameters for Part A are determined as $Z_1 = 91 \Omega$, and $\theta_1 = 39.7^\circ$, achieving a maximum frequency ratio of $k = 1.8$ and consequently an upper band edge of 4.5 GHz. As shown in Fig. 2(b), the imaginary part of Z_{in1} monotonically increases from a negative state to a positive one, and the zero-crossing point, labeled as f_0 , is 3.85 GHz.

B. Design of Part B

As shown in Fig. 1, Part B is an open-circuited stub with a characteristic impedance Z_2 and electric length θ_2 at f_L . It is a parallel type of resonance at $f_0 = 3.85$ GHz, and its admittance varies inversely when compared with the imaginary part of Y_{in1} . So, the imaginary part of the input admittance Y_{in1} around f_0 can be compressed with a parallel connection of Part B.

The electric length of Part B is $\lambda/2$ at f_0 , so θ_2 at f_L equals to $\pi f_L / f_0$. Then the input admittance of Part B can be expressed as

$$Y_{inB}(f) = \frac{j \tan(\theta_2 \cdot f / f_L)}{Z_2}. \quad (4)$$

As labeled in Fig. 1, the input admittance is denoted as Y_{in2} when Part B is connected in parallel with Part A, so Y_{in2} is expressed as

$$Y_{in2} = Y_{in1} + Y_{inB}. \quad (5)$$

As seen in Fig. 3(a), due to Y_{inB} is symmetrical with respect to f_0 and varies inversely when compared with the imaginary part of Y_{in2} around f_0 . When summed, their opposite-signed imaginary parts cancel at these frequencies, creating two equal points as f_{01} and f_{02} .

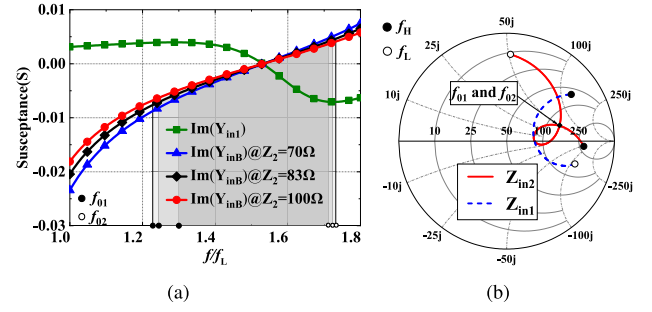


Fig. 3. (a) Y_{in1} over f_L – f_H and Y_{inB} with different Z_2 . (b) Z_{in1} for Part A and Z_{in2} after being compensated by Part B.

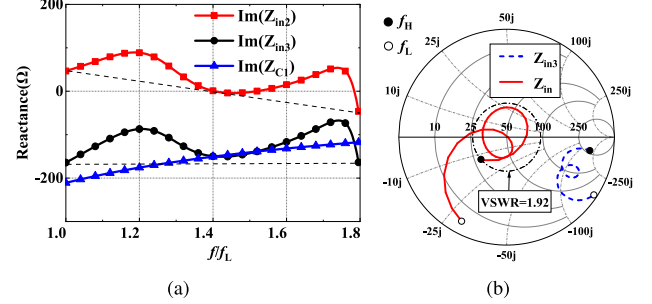


Fig. 4. (a) Imaginary part of Z_{in2} and the impedance of Part C over f_L to f_H . (b) Z_{in3} for the impedance after being compensated by Part C.

However, different Z_2 values yield varying slopes of Y_{inB} , thereby altering the frequency gap between f_{01} and f_{02} . To maximize the impedance compression around f_0 , a target optimization for frequency band $BW = f_{02} - f_{01}$ is performed by tuning the characteristic impedance Z_2 . The solution space is constrained by

$$Y_{in2}(f_{01}) = Y_{in2}(f_{02}); f_{01} < f_0 \text{ \& } f_{02} > f_0. \quad (6)$$

For verification, k was calculated with different Z_2 and θ_2 based on the physical limitations of PCB manufacturing. As shown in Fig. 3(a), it is evident that $Z_2 = 83 \Omega$ yields the widest gap between f_{01} and f_{02} , and the impedance trajectory over f_{01} and f_{02} is tightly clustered within a small region [Fig. 3(b)], indicating impedance compression at the middle part of the target band.

C. Design of Part C

As shown in Fig. 4(a), despite fluctuations, the overall trajectory of the imaginary part of Z_{in2} over f_L – f_H shows a net decrease from positive to negative impedance. To neutralize the variation in $\text{Im}(Z_{in2})$ with the increase in frequency, a capacitive reactance $-j/\omega C$ with an opposite tendency can be used.

Part C is a capacitor labeled as C_1 . After attaching C_1 to the circuit consisting of Part A and B, the input impedance is $Z_{in3} = Z_{in2} - j/\omega C_1$. Once being well-compensated by C_1 , the imaginary part of Z_{in3} over f_L – f_H is supposed to satisfy

$$\text{Im}[Z_{in3}(f_L)] = \text{Im}[Z_{in3}(f_H)]. \quad (7)$$

Thus, C_1 can be solved as

$$C_1 = \frac{f_H - f_L}{2\pi f_L f_H (\text{Im}[Z_{in2}(f_H)] - \text{Im}[Z_{in2}(f_L)])}. \quad (8)$$

In the design, C_1 was calculated as 0.302 pF with a given Z_{in2} , and Fig. 4(a) shows Z_{in3} after being compensated by C_1 . Compared with Fig. 3(b), the upper curve of Z_{in3} is bent toward

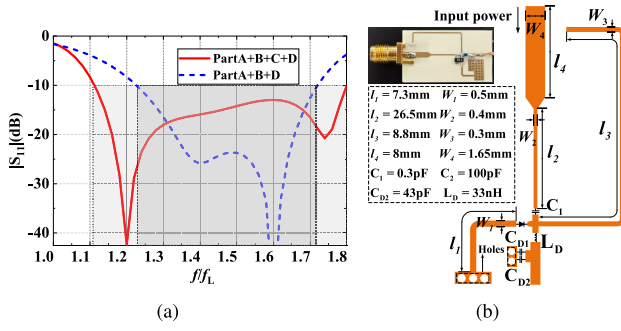


Fig. 5. (a) Calculated $|S_{11}|$ for broadband rectifiers with/without Part C at 0 dBm. (b) Layout, component values, and a photograph of the proposed rectifier.

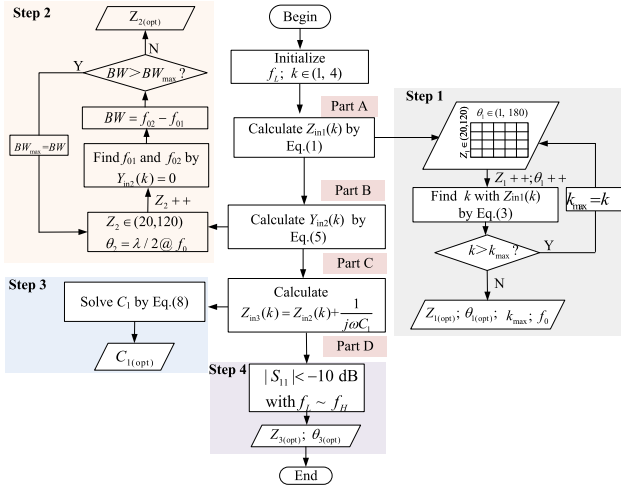


Fig. 6. Flowchart for the design of the proposed broadband rectifier.

the horizontal axis of the Smith chart, indicating a significant reduction in the imaginary part of the input impedance.

D. Design of Part D

As shown in Fig. 4(b), the trajectory of Z_{in3} has been concentrated in a small region. Part D, a section of transmission line with a high characteristic impedance, can be used to transform Z_{in3} to source impedance. In a broadband matching, Z_3 and θ_3 of Part D can be determined when the input impedance of the rectifier satisfies $|S_{11}| < -10$ dB within the targeted frequency range.

Fig. 4(b) also shows the trajectory of Z_{in} when Z_{in3} has been transformed by Part D along with $Z_3 = 120 \Omega$ and $\theta_3 = 39.5^\circ$. As can be seen, this curve is close to 50Ω on the Smith chart, indicating that matching is achieved in the frequency band from 2.75 to 4.5 GHz.

For a comparison to broadband matching techniques reported in [15] and [17], Fig. 5(a) shows $|S_{11}|$ of the proposed rectifier when compared with a reference broadband rectifier without using Part C. It can be seen that after being compensated by Part C, the bandwidth for $|S_{11}| < -10$ dB has been broadened from 3.1–4.3 to 2.75–4.5 GHz. The fractional bandwidth increases from 32.4% to 48.3%, representing an increase of 15.9%.

Based on the above discussions, the initial parameters for the proposed broadband design are identified. A flowchart showing the design procedure is illustrated in Fig. 6, which consists of four steps. In step 1, optimal Z_1 and θ_1 are obtained by maximizing k under the condition $\text{Im}[Z_{in1}(f_L)] =$

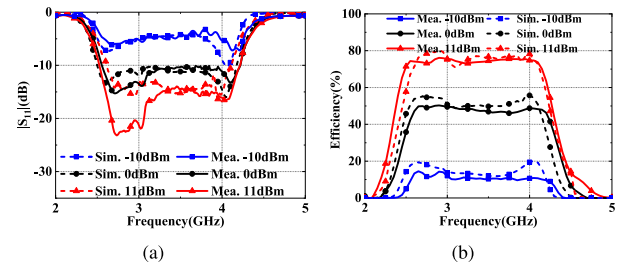


Fig. 7. Simulated and measured (a) $|S_{11}|$ and (b) PCE versus frequency. The dc load resistance is 500 Ω .

TABLE I
COMPARISON TO SOME PRIOR RECTIFIERS

Ref.	Efficiency over operating band		Matching Method	Diode, circuit area (λ^2 *)	Diode num.
	FBW (%)	Measured bandwidth (GHz)			
[14]	44.9%	0.57-0.9 > 70% @ 14.1 dBm	Parallel resonator	HSMS 2860 & 0.15x0.22	1
[17]	44.4%	2.1-3.3 > 70% @ 14 dBm	Parallel resonator	HSMS 2860 & 0.28x0.16	1
[18]	41.5%	2.0-3.05 > 70% @ 10 dBm	Multi-stage TL	HSMS 2860 & 0.3x0.29	1
[19]	48.9%	1.7-2.8 > 70% @ 11 dBm	Serial Coupled TL	HSMS 2860 & 0.28x0.08	1
This work	51.9%	2.5-4.25 > 70% @ 11 dBm	Parallel & Serial compensation	HSMS 2860 & 0.31x0.19	1

* λ is the guided wavelength at the center frequency of the band

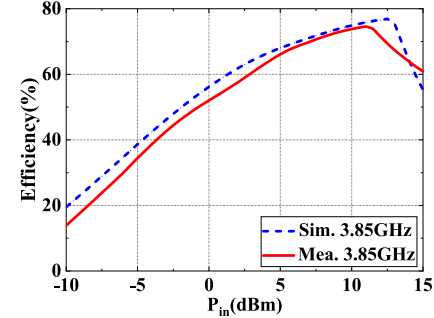


Fig. 8. Simulated and measured PCEs of the proposed rectifier versus input power level at 3.85 GHz.

$-\text{Im}[Z_{in1}(kf_L)]$. In step 2, Z_2 and θ_2 are calculated by maximizing the frequency range for reducing the impedance variation around f_0 . After that, in step 3, C_1 is computed by (8) for impedance compensation at the lower segment of the band $f_L - f_H$. Finally, Z_3 and θ_3 can be determined when the input impedance of the rectifier satisfies $|S_{11}| < -10$ dB within the targeted frequency range.

III. IMPLEMENTATION AND MEASUREMENT

The proposed rectifier is fabricated on a substrate of Rogers 4350B ($\epsilon_r = 3.66$ and $\tan \delta = 0.002$) with a height of 0.76 mm. The layout and the dimensions of the rectifier are shown in Fig. 5(b), and a photograph of the fabricated circuit is also shown. The dc-pass filter with a broad stopband is composed of C_{L1} , C_{L2} , and L_D . When R_L is relatively large ($> 100 \Omega$), the cutoff frequency of this low-pass filter, f_{cut} , can be estimated by $1/2\pi(L_D(C_{L1} + C_{L2}))^{1/2}$. Fine optimizations are carried out using the Advanced Design System (ADS), Keysight.

In Fig. 7(a), the simulated and measured $|S_{11}|$ at the input power of 11, 0, and -10 dBm are shown. At 11 dBm, the measured $|S_{11}|$ is less than -10 dB from 2.5 to 4.25 GHz, which corresponds to a fractional frequency bandwidth of 51.9%. The bandwidth for $|S_{11}| < -10$ dB at 0 dBm is the same as that of 11 dBm. Besides, the measured power conversion efficiency (PCE) versus frequency at 11 and 0 dBm are plotted in Fig. 7(b), together with the simulation results. As shown, an efficiency of higher than 70% can be achieved from 2.5 to 4.25 GHz, which corresponds to a fractional bandwidth of 51.9%. Meanwhile, in this frequency range, the measured efficiency is also maintained at more than 45% when the input power is 0 dBm. The measured results agree with the simulated ones.

Fig. 8 presents the simulated and measured PCEs versus input power at 3.85 GHz, which corresponds to the center frequency of the 2.5–4.25-GHz band. As observed, the measured PCE of the proposed design is greater than 60% over 3–15 dBm, with a peak PCE of 74.6% obtained at 11 dBm.

Table I shows a comparison of the performances between the proposed rectifier and the works featuring broadband in the literature. As shown, the proposed rectifier shows a fractional bandwidth of 51.9% for the PCE higher than 70%, which is the highest among all the designs under comparison. The physical size is medium compared with other designs.

IV. CONCLUSION

In this letter, a novel broadband matching network is proposed using shunt and serial impedance compensation. After being compensated by a parallel resonator, the input impedance of the rectifier can be further compressed by a serial capacitance. The design theory of the matching network has been analyzed, and the closed-form design formulas have been derived, so that circuit parameters can be calculated once the start frequency of the targeted band is given. Compared with a broadband matching using only one parallel resonator for impedance compensation, the proposed design improves the bandwidth for $|S_{11}| < -10$ dB by 15.9%. For validation, a rectifier operating at 2.5 – 4.25 GHz was fabricated and measured. The measurement results show a wide bandwidth of 51.9% when the efficiency is more than 70%.

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